

Portfolio News Monitoring & Summarization

Technical Choices

Models:

- Teacher model: google/flan-t5-base (used for baseline summarization)
- Student model: Same base (flan-t5-base) fine-tuned using LoRA
- Evaluation model: facebook/bart-large-mnli used for factuality checking

Fine-Tuning Strategy:

- Used PEFT (LoRA) to adapt Flan-T5 without full weight updates
- Fine-tuned for 3 epochs, with batch size 8 and learning rate 1e-4
- Input truncation at 512 tokens, output truncation at 150 tokens

Data Sources:

- Original CSV: Financial_News_Dataset.csv. The dataset is fully synthetic and it contains 4000 data points.
- Fields used: text, summary, stock, url, timestamp
- Processed into JSONL format for training, inference and evaluation

Example of the first two datapoints of the dataset

stock	text	summary	timestamp	URL
NVDA	NVDA issued a downward revision of its earnings forecast due to unexpected headwinds in supply chains. This marks the second time in the fiscal year that the company has adjusted its guidance. Shares fell sharply following the announcement, with analysts expressing concerns over long-term profitability.	NVDA issued a downward revision of its earnings forecast due to unexpected headwinds in supply chains — a move that has drawn attention across Wall Street.	2025-07-10 12:53:13	https://www.russell.net/news/north-under-choice
JPM	JPM faced regulatory scrutiny over its data privacy practices, prompting a formal investigation. While the company has pledged	Headline: JPM faced regulatory scrutiny over its data privacy practices,	2023-09-03 17:50:02	https://www.cherry.net/news/business-building

	full cooperation, shares dipped slightly amid investor uncertainty. Analysts are watching closely for any implications on future earnings.	prompting a formal investigation.		
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Reasoning Behind Choices

The choice of using **google/flan-t5-base** as the base model came from its strong performance on a variety of summarisation tasks. Flan-T5 is a variant of T5 that has been instruction tuned, making it effective in zero-shot and few-shot scenarios which is an important factor since financial news can vary in structure and complexity. It gave a good starting point for both generating strong baseline summaries and fine-tuning a model.

To adapt Flan-T5 to this domain without incurring the high computational cost of full fine-tuning, **LoRA** was used. **LoRA** injects a small number of trainable parameters into specific layers of the model, allowing to fine-tune with fewer GPU resources and less training time. This choice was important for working within the compute limits of Google Colab, even with access to an A100 GPU, which was used.

For the evaluation side, a few metrics were calculated to get a good picture of the summary quality. **ROUGE-L** was used to assess textual overlap with the original content and **BERTScore** to evaluate semantic similarity, both those metrics are ideal for summarisation. Also, the summaries were tested to see whether they were factually grounded in the input articles. To do this, the **facebook/bart-large-mnli** was used, a model trained for natural language inference. It allowed to judge whether a generated summary was entailed by the source text, therefore giving a way to flag hallucinations. This was achieved by calculating the entailment score of a summary and if it was below 0.5 it was saved as a hallucination. Using this external NLI model as a factuality checker was a practical and scalable way to quantify something that's otherwise very subjective.

In terms of data processing, a raw CSV containing financial articles and summaries was developed synthetically using an LLM agent after some failed tries to get some data using a scraper and trying to find an ideal dataset for this scenario. Some malformed entries were filtered out and minimal text cleaning was applied. Summaries under 6 words were discarded to prevent the model from learning from too simple examples. Truncation settings for token limits (512 for inputs, 150 for outputs) were chosen to balance completeness and model constraints.

Challenges and Solutions

One of the challenges faced was getting the pipeline to train end-to-end. At first the fine-tuning loop logged a flat **loss = 0.0** which turns out that every label token was masked to -100 because I'd zipped against the wrong attention mask, so the model literally had nothing to learn. After spotting that, I fixed the masking logic and loss started dropping normally.

I also hit NaN gradients once I switched on fp16, lowering the learning rate and disabling mixed precision for this small model solved it.

Another issue that was faced, was that the Docker image kept crashing because the image tried to pull heavy CUDA dependencies. So instead of using GPU drivers, I switched to **torch 2.2.2+cpu** in requirements.txt to use CPU only PyTorch to avoid downloading large CUDA dependencies during Docker build and removed the **deploy_gpu** from the docker-compose.yml file.

Finally, a bug in the metrics script, calling **calculate_bertscore** as a global instead of a method gave me strings of nan until I refactored it.

Evaluation results and discussion

Evaluation Table:

Metric	Baseline	Fine-Tuned
ROUGE-L	0.454	0.507
BERTScore	0.758	0.758
Factuality Score	0.441	0.568

The evaluation shows that fine-tuning with LoRA brought some improvements. ROUGE-L rose from 0.454 to 0.507, which means the fine-tuned summaries better matched the original article content in terms of key phrases and structure. BERTScore stayed the same at 0.758. This suggests that even before tuning, Flan-T5 was generating semantically relevant summaries, but fine-tuning didn't change that.

The most important shift came in factuality. The entailment score jumped from 44.1% to 56.8%, showing that the fine-tuned model produced summaries that were better grounded in the source article. In other words, it hallucinated less. Overall, these results show the fine-tuned model is not just repeating facts more clearly but is also more reliable.

Hallucination Example

Type: baseline_summary

Entailment Score: 0.309

Original Text:

“NVDA issued a downward revision of its earnings forecast due to unexpected headwinds in supply chains. This marks the second time in the fiscal year that the company has adjusted its guidance. Shares fell sharply following the announcement, with analysts expressing concerns over long-term profitability.”

Generated Summary (Hallucination):

"NVDA has lowered its earnings forecast for the first time in the fiscal year, citing a weakening of supply chains."

Why it's a hallucination:

The summary claims it was the "first time" NVDA lowered its forecast, but the article clearly states it was the **second time**.

Examples of the summarised news

Example 1 — NVDA

Original Text:

“NVDA issued a downward revision of its earnings forecast due to unexpected headwinds in supply chains. This marks the second time in the fiscal year that the company has adjusted its guidance. Shares fell sharply following the announcement, with analysts expressing concerns over long-term profitability.”

Baseline Summary:

"NVDA has lowered its earnings forecast for the first time in the fiscal year, citing a weakening of supply chains."

Fine-tuned Summary:

"Headline: NVDA issued a downward revision of its earnings forecast due to unexpected headwinds in supply chains."

Example 2 — JPM

Original Text:

“JPM faced regulatory scrutiny over its data privacy practices, prompting a formal investigation. While the company has pledged full cooperation, shares dipped slightly amid investor uncertainty. Analysts are watching closely for any implications on future earnings.”

Baseline Summary:

"Shares in JPMorgan Chase & Co. slipped after the company faced regulatory scrutiny over its data privacy practices, prompting a formal investigation by regulators."

Fine-tuned Summary:

"Investor takeaway: JPM faced regulatory scrutiny over its data privacy practices, prompting a formal investigation."

Example 3 — AAPL

Original Text:

“AAPL faced regulatory scrutiny over its data privacy practices, prompting a formal investigation. While the company has pledged full cooperation, shares dipped slightly amid investor uncertainty. Analysts are watching closely for any implications on future earnings.”

Baseline Summary:

"AAPL's shares dipped slightly after a formal investigation into its data privacy practices, prompting an investigation into the company's practices."

Fine-tuned Summary:

"Investor takeaway: AAPL faced regulatory scrutiny over its data privacy practices, prompting a formal investigation."

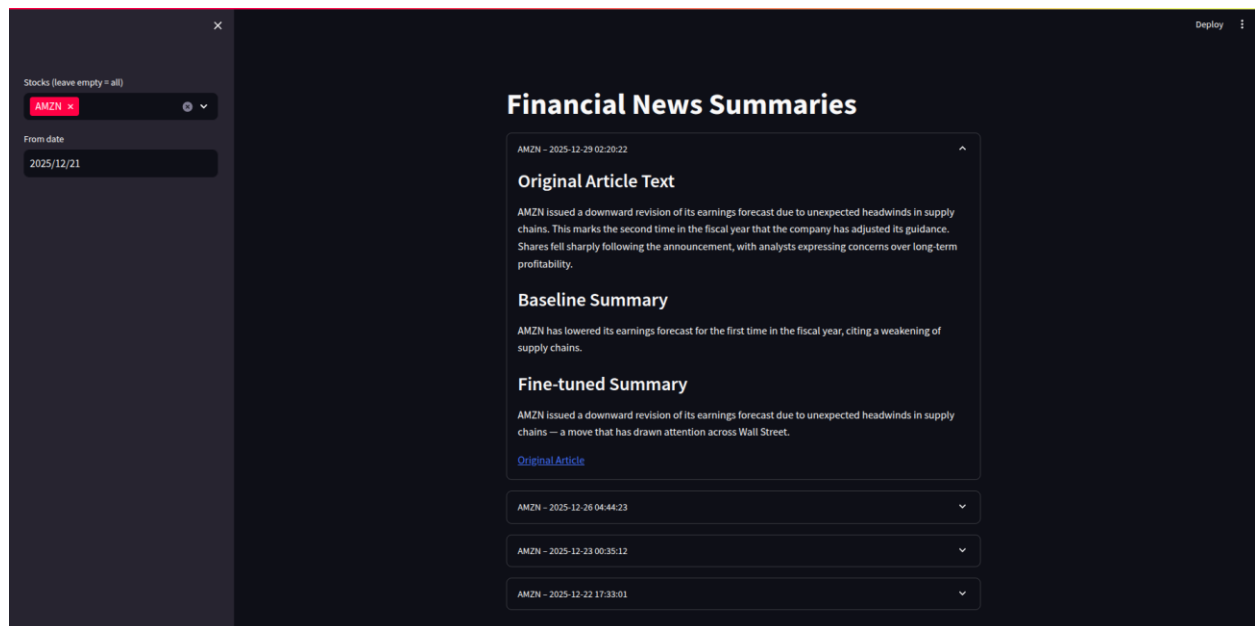


Figure 1: UI Showcase